STATEMENT OF WORK

MATLAB

Background:

Sponsor / Customer: NAVSEA, PMS 408 Expeditionary Missions

End User: Navy (CIED/CUAS), Air Force, FMS – Australia

Requirement: The Software Support Activity (SSA) provides a full software sustainment capability for the JCREW I1B1 System of Systems. The goal is to support / lead system integration and technical insertion efforts as directed by the I1B1 APM/PM. The SSA will implement the system software and firmware (SW/FW) supportability and sustainment requirements identified in the Capabilities Production Document (CPD) and the Life Cycle Sustainment Plan (LCSP).

Project Completion: The JCREW I1B1 is a **DoD Program of Record (POR) - ACAT II** program with major milestones through its life cycle where SSA has a direct involvement.

NSWC IHD:

RDT&E VSYSCOM Lab 209, E34, **NSWC IHD**, serves PMS408 - Joint Counter Radio-Controlled Improvised Explosive Device Electronic Warfare (JCREW) as the **Software Support Activity** (SSA) for JCREW I1B1 Family of Scanner/Jammers. The nature of the software/firmware makes it imperative that the **SSA** must "mirror" the software development environment of the prime vendor, Northrup Grumman for sustainment (main deliverable software/firmware), transformational (NEXTGEN SDR, SC, CDU), and support of the JCREW Organic Depot at NSWC CN.

The JCREW family of systems in deployed to the US NAVY, US AIRFORCE, OGA, and FMS customers.

The JCREW SSA must handle SW from other USG stakeholders, JHU/APL, and Industry.

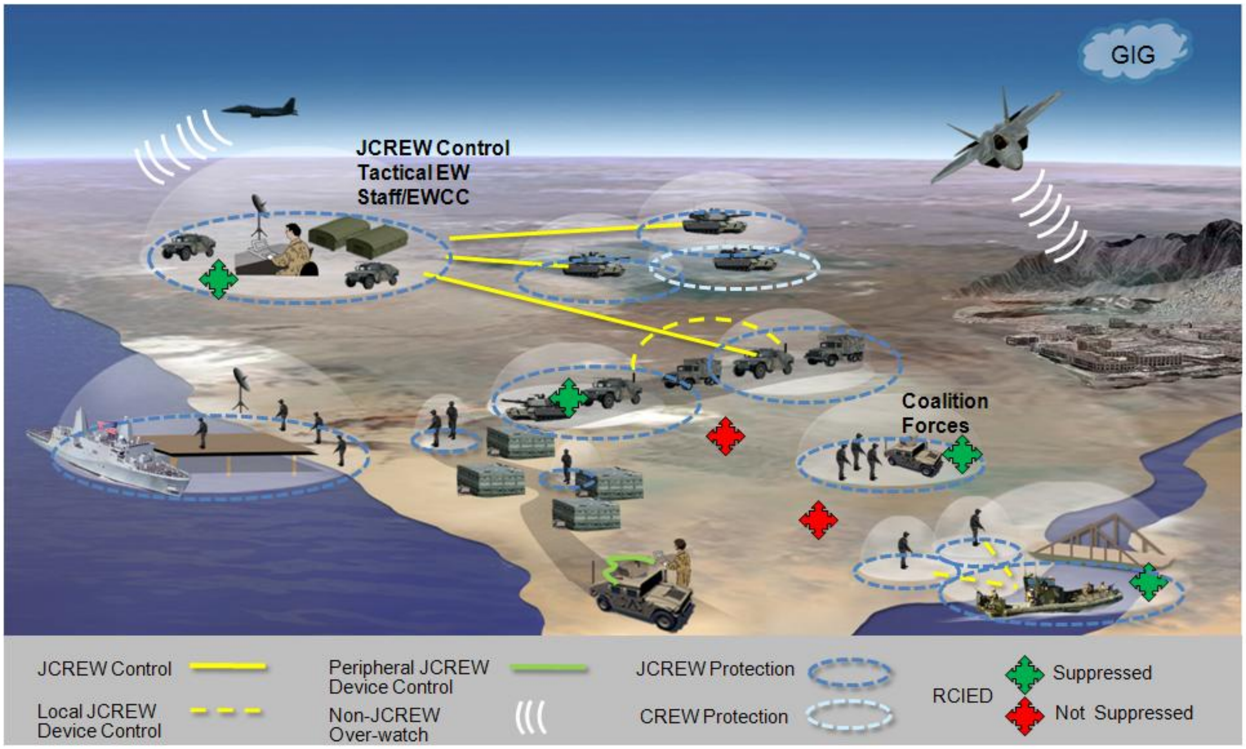
This document is prepared in accordance with the direction of the Joint Counter Radio Controlled Improvised Explosive Device Electronic Warfare (JCREW) program Statement of Work (SOW).

Identification:

This Computer Software Product (CSP) describes how to build each of the JCREW software and Very High Speed Integrated Circuit (VHSIC) Hardware Description Language (VHDL) components and the Digital Versatile Discs (DVDs) that contain all the source code, test and peer review artifacts for the CSCIs.

System Overview:

JCREW System of Systems (SoS), shown in shown below, implements a network-ready system architecture that is to be integrated into a variety of host platforms (dismounted, mounted and fixed site/semi-permanent). JCREW Devices share a common modular hardware and software design for a cost effective solution using electronic warfare (EW) methods and techniques that meets the war fighter’s need for an agile and adaptive active defense system to suppress ever changing Radio Controlled Improvised Explosive Devices (CIEDs) AND (CUAS) threats.



MATLAB is needed in support of NEXTGEN SDR/SC, CDU 2.0, FOA, DRAKE, QUATTRO/MIB technical instructions -- high priority CUAS/CIED projects.

Need:

JCREW SSA requires MATLAB license RENEWALS. JHU/APL shares its TI work to NGC, NSWC CN, and SSA via MATLAB files. Increased effort is being made in including Machine Learning/AI into the future JCREW SW upgrades and MATLAB is a vital component.

MATLAB has been selected by the Prime Vendor, Northrop Grumman Corp., to develop the Next Generation of JCREW assemblies.

MATLAB (an abbreviation of "MATrix LABoratory") is a proprietary multi-paradigm programming language and numeric computing environment developed by MathWorks. MATLAB allows matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, and interfacing with programs written in other languages.

Although MATLAB is intended primarily for numeric computing, an optional toolbox uses the MuPAD symbolic engine allowing access to symbolic computing abilities. An additional package, Simulink, adds graphical multi-domain simulation and model-based design for dynamic and embedded systems.

MATLAB combines a desktop environment tuned for iterative analysis and design processes with a programming language that expresses matrix and array mathematics directly. It includes the Live Editor for creating scripts that combine code, output, and formatted text in an executable notebook, is needed in support of TI NEXTGEN SDR, CDU 2.0, SC 2.0, FOA, DRAKE, QUATTRO/MIB technical instructions -- high priority CUAS/CIED projects as well as the C-IED threats that are addressed by the FPGA, CPLD, and SoC in current and future JCREW systems. JCREW SSA needs to maintain their existing MATLAB tool base, and expand with new tools.

MatLab (~~New Products~~Modules)

- **5G Toolbox (5G)**
    - 5G Toolbox™ provides standard-compliant functions and reference examples for the modeling, simulation, and verification of 5G New Radio (NR) communications systems. The toolbox supports link-level simulation, golden reference verification, conformance testing, and test waveform generation. With the toolbox you can configure, simulate, measure, and analyze end-to-end 5G NR communications links. You can modify or customize the toolbox functions and use them as reference models for implementing 5G systems and devices. The toolbox provides functions and reference examples to help you characterize uplink and downlink baseband specifications and simulate the effects of RF designs and interference sources on system performance. You can generate waveforms and customize test benches, either programmatically or interactively using the Wireless Waveform Generator app. With these waveforms, you can verify that your designs, prototypes, and implementations comply with the 3GPP 5G NR specifications.
- **Filter Design HDL Coder (FH)** - Filter Design HDL Coder™ generates synthesizable, portable VHDL® and Verilog® code for implementing fixed-point filters designed with MATLAB® on FPGAs or ASICs. It automatically creates VHDL and Verilog test benches for simulating, testing, and verifying the generated code.
    - The design entry input to Filter Design HDL Coder is a quantized filter that you create in one of two ways:
        - Design and quantize the filter with DSP System Toolbox
        - Design the filter with Signal Processing Toolbox™ and then quantize it with DSP System Toolbox
    - Filter Design HDL Coder supports several important filter structures, including:
        - Discrete-time finite impulse response (FIR), which includes symmetric, anti-symmetric, and transposed structures
        - Second-order section (SOS) infinite impulse response (IIR), which includes direct form I, II, and transposed structures
        - Multirate filters, which includes cascaded integrator-comb (CIC) interpolator and decimator, direct-form FIR and transposed FIR polyphase interpolator and

        - decimator, FIR hold and linear interpolator, and FIR polyphase sample rate converter structures
        - Fractional delay filters, which includes Farrow structures
    - Filter Design HDL Coder can generate HDL code from cascaded multirate and discrete-time filters. Each of these single-rate and multirate filter structures supports fixed-point and floating-point (double precision) realizations. In addition, the FIR structures support unsigned fixed-point coefficients.
- **Mapping Toolbox (MG)** - Mapping Toolbox™ provides algorithms and functions for transforming geographic data and creating map displays. Visualize your data in a geographic context, build map displays from more than 60 map projections, and transform data from a variety of sources into a consistent geographic coordinate system. Mapping Toolbox supports a complete workflow for managing geographic data. Import vector and raster data from a wide range of file formats and web map servers. The toolbox processes and customizes data using trimming, interpolation, resampling, coordinate transformations, and other techniques. Data can be combined with base map layers from multiple sources in a single map display. Export data in file formats such as shapefile, GeoTIFF, and KML.
- **Wireless HDL Toolbox (LH)** - Design and implement 5G and LTE communications subsystems for FPGAs, ASICs, and SoCs
    - Wireless HDL Toolbox™ provides pre-verified, hardware-ready Simulink® blocks and subsystems for developing 5G, LTE, and custom OFDM-based wireless communication applications. It includes reference applications, IP blocks, and gateways between frame- and sample-based processing.
    - Modify the reference applications for integration into design. HDL implementations of the toolbox algorithms are optimized for efficient resource usage and performance for prototyping or for production deployment on FPGA, ASIC, and SoC devices.
    - The toolbox algorithms are designed to generate readable, synthesizable code in VHDL® and Verilog® (with HDL Coder™). For over-the-air testing of 5G, LTE, and custom OFDM-based designs, you can connect transmitter and receiver models to radio devices (with Communications Toolbox™ hardware support packages).

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- MATLAB (MLSMS)
- Simulink (SLSMS)
- 5G Toolbox (5GSMS)
- Communications Toolbox (CMSMS)
- DSP System Toolbox (DSSMS)
- Embedded Coder (ECSMS)
- Filter Design HDL Coder (FHSMS)
- Fixed-Point Designer (POSMS)
- HDL Coder (HDSMS)
- Instrument Control Toolbox (ICSMS)

- MATLAB Coder (MESMS)
- MATLAB Compiler (COSMS)
- MATLAB Compiler SDK (MJSMS)
- Mapping Toolbox (MGSMS)
- RF Blockset (RBSMS)
- RF Toolbox (RFSMS)
- Signal Processing Toolbox (SGSMS)
- Simscape (SSSMS)
- Simulink Coder (RTSMS)
- Wireless HDL Toolbox (LHSMS)

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- MATLAB (MLSMS)
- Communications Toolbox (CMSMS)
- DSP System Toolbox (DSSMS)
- Embedded Coder (ECSMS)
- Fixed-Point Designer (POSMS)
- HDL Coder (HDSMS)
- Instrument Control Toolbox (ICSMS)
- MATLAB Coder (MESMS)
- MATLAB Compiler (COSMS)
- MATLAB Compiler SDK (MJSMS)
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- Curve Fitting Toolbox (CFSMS)
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- Signal Processing Toolbox (SGSMS)

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- MATLAB Compiler (COSMS)

MatLab (New Product) Individual License

- MATLAB (MLSMS)
- Simulink (SL)
- MATLAB Report Generator (MR)
- Optimization Toolbox (OP)
- Signal Processing Toolbox (SG)

MatLab (New Product) Individual License

- MATLAB (MLSMS)
- MATLAB Compiler (COSMS)

MatLab (New Product) Individual License

- MATLAB (MLSMS)

MatLab (New Product) Concurrent Licenses

- 4x MATLAB (MLSMS)
- 2x Simulink (SL)
- 2x DSP System Toolbox (DS)
- 2x Signal Processing Toolbox (SG)
- Simscape (SSSMS)

MATLAB® develops algorithms much faster than in traditional languages such as C, C++, or Fortran. The JCREW SSA can validate concepts, explore design alternatives, and distribute algorithms in the form that best suits the JCREW application. MATLAB provides the tools you need to transform ideas into algorithms, including:

- Thousands of core mathematical, engineering, and scientific functions
- Application-specific algorithms in domains such as signal and image processing, control design, computational finance, and computational biology
- Development tools for editing, debugging, and optimizing algorithms
- These capabilities, combined with MATLAB programs created by the worldwide user community, explores approaches that otherwise would be too time-consuming to consider.